

Revisiting Distance Learning Resources for Undergraduate Research and Lab Activities during COVID-19 Pandemic

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ABSTRACT: The rapid transition in teaching format due to COVID-19 gives rise to many challenges, especially for research and lab activities since they may require a different approach with more constrained resources compared with lecture-based courses, in which various virtual communication platforms have now been employed. As essential parts for student's active learning, effective strategies to perform undergraduate research and lab courses during the current circumstance need to be carefully designed. To address this challenge, we have revisited different distance learning resources and implemented four critical methods in undergraduate research activity in our research groups, including question-driven literature review, visualizing experiments from virtual scientific resources, performing safe and simple home-lab experiments, and learning new computational tools. These approaches provide versatile opportunities for remotely engaging students in research and lab activities besides just participation in group meetings and scientific webinars. We believe insights gained from revisiting and implementing these resources could have a long-term positive impact for improving teaching and mentoring infrastructure for undergraduate lab activities post-COVID-19.

KEYWORDS: General Public, Laboratory Instruction, Analogies/Transfer, Distance Learning/Self Instruction, Laboratory Computing/Interfacing, Undergraduate Research

Revisiting distance learning resources



BACKGROUND

The outbreak of COVID-19, a highly contagious respiratory disease, changes the way we live in every aspect.¹ Most universities have taken safety precautions in accordance with the stay-at-home order in order to mitigate the spread. Consequently, all courses were required to immediately transit from face-to-face instructions to an online format within a very short period of time during the ongoing pandemic. While in many universities distance learning has already been implemented into some degree programs, it was rarely used for chemistry-related curriculum, which often requires a dynamic learning environment and hands-on laboratory activities. Nevertheless, instructors have tried to maintain high teaching quality using various virtual platforms for class and office hours. Key benefits from online teaching that have been recognized include the ability of students to access prerecorded lectures and notes at any time to grasp concepts in a more efficient manner, as well as the potential of enabling open access to course content among different institutions.² Meanwhile, there are many challenges for remotely delivering classes in addition to the possible resource-constrained situations. It is apparent that using online platforms often leads to a much less interactive environment compared with in-person communications that could reduce the level of student

engagement. In the current situation, group learning and related activities that develop students' perspectives and give support from peers to build academic confidence become more difficult for students to execute. Furthermore, student learning amid the pandemic also involves an increased level of anxiety from the concerns about their future career, especially considering that the current unemployment rate is high.³ As educators, we need to be more thoughtful to not only help students gain and practice new knowledge during this unprecedented time, but also understand their difficulties and challenges. One approach that we have applied is to create a less stressful environment with a slightly lighter load of assignments, but at the same time using leading questions in homework to initiate a self-learning process for new knowledge, which improves teaching productivity in subsequent classes since students have already previewed the content to a certain degree. While efforts have been made from both

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students and teachers to rapidly adapt to the current situation, the effectiveness of different virtual approaches for active learning still needs careful validation. More importantly, feedback from students will provide important insights for us to improve the quality of online teaching in the future.

Furthermore, undergraduate research and laboratory classes, important platforms for STEM students to translate fundamental knowledge into practical skills, are now more constrained by campus closures compared with lecture-based courses. As an essential piece for student development, participation in research projects motivates students to learn problem-solving and improve scientific communication skills in order to be prepared for confidently addressing future challenges in their careers and life. However, the significant slowdown of research and lab activities leads to an enormous negative impact on the ultimate learning outcome of undergraduate study in chemistry and other related majors. From a student's perspective, these lab activities are fun experiences for building up professional skills and inspiring them to further their educations. Many students expressed the concerns that they missed working in the lab after schools announced campus closure. Therefore, it is important to swiftly find a solution to provide best-available resources for students to remotely engage in research activities in an efficient manner. Herein, we first discuss the challenges of student research and lab activities that emerged after the rapid transition to online format and then summarize key methods that we have employed in the research groups to engage undergraduate research such as question-driven literature review and visualizing experiments from virtual journals. While performing experiments at home is not an ideal scenario, some safe projects would be useful to inspire students to connect textbook knowledge with chemistry in everyday life. Finally, data analysis and using a computational approach are emerging areas where students can develop the necessary skills in this high-tech era. We hope to utilize these experiences and insights gained during COVID-19 to enhance the infrastructure of research mentoring and lab courses in the postpandemic future.

■ CHALLENGES

Undergraduate research and lab courses are important parts of chemical education, and these "learning by doing" activities usually require the use of equipment and chemicals that are readily available in the laboratory. COVID-19 caused us all to hit the "pause" button on in-lab activities, and their transition to distance learning is inevitably more difficult than class lectures. During the first 2 weeks after transition, we tried to maintain undergraduate research by engaging students in research group meetings and scientific webinars. While students show engagement in these activities, we soon realized they do not provide the complete experience that students should receive from undergraduate research. This could lead to potential loss of interest from the students due to the lack of practical engagement; thus, managing the students' disappointment and anxiety from not being able to conduct research is also of great importance. We utilized this opportunity to teach the students that rapid changing (or even totally expected) situations sometimes can happen during their professional career, so they have to be able to adapt accordingly and quickly. We have struggled to seek better solutions to provide students more comprehensive training during their stay at home so they can harvest the benefits of good career

preparation and professional development. For example, students are encouraged to explore a relatively new research area for broadening their scope of knowledge for potential interdisciplinary research projects. The urgent need for a swift adaptation demands that we revisit all available resources for distance learning that can be implemented into undergraduate research and lab courses. During this revisiting process, we kept the following questions in mind in order to provide the best possible learning experience and outcome for students.

- Can we direct student-learning process to be more productive and efficient?
- What hands-on practices can be employed during the current situation?
- Can we use the insights gained during COVID-19 to further improve future teaching infrastructure?
- Can we utilize the current situation to develop students' self-learning ability? and if so, what are the key things they need to learn?

While the rapid switch to remote learning has been challenging for both students and instructors during this special time, we hope that these distance learning resources and the experiences we encountered can be implemented into the infrastructure of chemical education post-COVID-19 to further improve the dynamic learning environment, as we summarized in Figure 1.

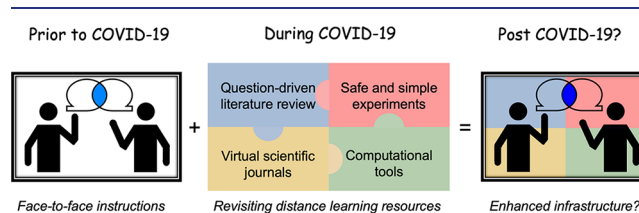


Figure 1. A scheme to show the possible enhanced infrastructure for undergraduate research and lab activities after COVID-19 by combining insights gained from revisiting distance learning resources with normal face-to-face instructions.

■ REMOTE LEARNING APPROACHES AND RESOURCES

Question-Driven Literature Review

Literature review builds up sufficient background knowledge for students to conduct research as the first step when entering a new field. We guided this process by asking central questions instead of just providing a few keywords for students to search in various databases.⁴ In brief, the question should be specific to lead the literature search and reading, and allow the ability to build up further questions for making continuous progress to broaden the knowledge scope. An example of a good research question would be "What are the main factors to consider for fabricating a polymer solar cell device with high efficiency?" This question involves a broad research topic as well as a specific section that students can address without entering intricate details. Writing a summary of their literature findings for answering the questions is useful for students to organize these new pieces of knowledge in a logical order, which teachers can then provide additional inputs to complement when necessary. After a discussion with students, an instructor may add a follow-up question on the topic, such as "In recent years, what levels of efficiency have been achieved successfully?" or "What has proven to be the biggest challenge

with these solar cells?" These follow-up questions not only help students to identify the remaining challenges in a research field, but also inspire them to address these technology/scientific knowledge gaps by performing research projects. On the other hand, a much less ideal situation would be instructor initiating a research project by only sending a few papers to students and asking them to study "polymer solar cell" and learn "how do they work?". While these are still legitimate questions, a new undergraduate researcher can sometimes dive too deep (and often unnecessarily) to find the answer and easily get lost in some difficult quantum chemistry after spending valuable time reading the literature. In our opinion, it is important to keep students' interest when they enter a new research field and mentor them according to their knowledge background. Overall, we found that this central question-driven literature review method is effective to engage students and improves their productivity during literature review to understand a new research topic by minimizing their confusion from massive works of literature. More importantly, students can generalize this approach for self-directed learning in the future when delving into further work.

Visualizing Experiments from Virtual Journals

Prior to the pandemic, many scientific journals already established platforms to publish a virtual experimental section, in which authors can upload a video to show how the research was performed. This now provides a great resource for students to engage in research activities from home through the observation of an experiment. In the virtual experimental sections, all steps can be followed closely, instead of solely relying on understanding the text. One of the platforms we used is through the Journal of Visualized Experiments, containing a step-by-step protocol section to elaborate the experimental procedure, accompanied by a video recorded and a voiceover for describing each step in a careful and precise manner.³ These resources facilitate the students' grasp of new chemistry techniques such as thin-film chromatography and column separation without accessing the lab.^{5,6} With the recent phased reopening of research laboratories, we are now inspired to use a portable GoPro camera to record electronically our experimental procedures for distance mentoring of undergraduate research. Students have indicated that they enjoyed these virtual experiences of "doing" experiments, as the ability to observe a procedure limits confusion and they are able to follow through an experiment and understand what to expect after each step. These virtual resources also bring benefits to outreach activities by making them available as showcases to K-12 students for remotely learning chemistry and being involved in research projects. The current situation not only highlights the importance of such visuals, but also may push them to be used more frequently as a critical platform in the future.

Safe and Simple Projects at Home

Chemistry is present in all aspects of our everyday life, but its role might be overlooked if students think they can only perform experiments in a lab. During this time when students do not have access to the lab, performing safe experiments at home is useful for drawing connections between chemistry and some common everyday items. The most important thing here is that the safety of each step needs to be carefully considered and examined by professionals and no toxic or hazardous chemicals should be involved.⁷ Here, we list a few projects that are suitable for home-lab research, including but not limited to, using a USB microscope to observe the expansion of dried

contact lens in water (polymer hydrogel swelling), heating and freezing commercial rubber bands or thermoplastic elastomers to make shape-memory materials,⁸ preparing hand sanitizer with rubbing alcohol and glycerin, and adjusting the freezing (crystallization) temperature of water by the addition of salts.⁹

We asked students to think independently to develop a step-by-step procedure for instructor's approval. We then provided feedback on these proposed procedures and asked students to report their observations from performing these experiments, followed by interactive students-instructor/mentor discussions. Observation from these simple and interesting experiments helped students better understand the knowledge learned in core chemistry courses. Moreover, it could be useful to ask students to identify and design control experiments, as understanding the significance of performing controls for validating hypotheses is essential for research projects. An important feedback we received from students was that performing safe and simple experiments from home helped decrease their level of anxiety and mitigated feelings of "not doing anything". Teachers can also use this approach to generate interest for students who are new to the chemistry field, such as implementing them into Introduction to Chemistry and other similar courses, as an important strategy for engaging them and preventing potential talent loss. While performing safe experiments at home seems to be a short-term solution, future community outreach can leverage these experiences to provide a compelling outlook on the role of chemistry in life for students at different ages.

Learning and Using Computational Tools

Simulation and computational tools are important in the chemistry field for understanding, predicting, and designing materials at multilength scales ranging from a molecular-level structure to a macroscopic property. While learning to operate a new simulation package from the ground up could be difficult, the high versatility of various software applications (open source or academic licensed) enables students to choose a suitable platform according to their needs and background. For example, both Python and MATLAB are well-suited for data interpolation and visualization, stochastic simulations, image analysis, and automated bulk data processing, and interested readers can find several excellent tutorials on related topics.^{10–12} There are also many web-based resources for more advanced molecular dynamics and Langevin dynamics simulation such as the Chemistry at Harvard Macromolecular Mechanics interface and graphics (CHARMMing),¹³ with great potential to facilitate students' understanding of theoretical principles of chemistry and relevant courses. ChemDraw and SOCOT (structure-based organic chemistry online tutorials) can be used to predict reaction products and their physical properties, as well as identifying reaction intermediates, aligning with some particular learning goals of core chemistry courses.¹⁰ Students with major interest in learning/using computational tools responded positively to this activity, but a few others faced a more difficult time due to the learning curve of some of these complicated programs. However, we believe this can be overcome through identifying a more suitable software since each program would provide a different advantage for experimental observation and simulation. With these practices, students can expand their capabilities by adding a new tool to enable a more comprehensive approach to solve problems in the future.

■ CONCLUSIONS

The current situation has imposed many constraints in teaching chemistry, but advances in information technology and the internet allow instructors to maintain a high quality of teaching by employing various online platforms. However, it is very difficult to transit lab-related activities to an online format. We recognize the challenges of performing undergraduate research and lab courses amid COVID-19, which could lead to potential loss of interest from students, since hands-on lab activities represent a vital practical aspect in the chemistry field. Four major approaches have been implemented in the research groups to enhance student engagement and learning outcomes by making the learning from home experience more efficient and fruitful. While the current situation challenges conventional approaches in student education, it also provides a good opportunity for instructors to revisit distance learning resources for teaching class and mentoring research. We believe these virtual resources and insights gained can be utilized to improve the infrastructure of the dynamic learning environment for chemical education post-COVID-19.

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Notes

The authors declare no competing financial interest.

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